CommonsLens: Automatic Category Suggestion for Wikimedia Commons

TEAM-89
Hackathon Problem Statement 3

1. Problem and Our Understanding

The objective is to automatically suggest relevant categories for newly uploaded Wikimedia Commons images, reducing manual categorisation backlogs and making media easier to search and reuse. We treat this as extreme multi-label classification over a continuously growing graph of millions of categories. Historical labels are volunteer-assigned, so they are often incomplete and inconsistent; editors typically tag only the most specific category, leaving broader ancestors implicit. Given the high daily upload volume and diverse metadata (titles, descriptions, EXIF, and geotags), the system must scale. Its output is a reliable suggestion to assist human review, never an automatic, unsupervised edit.

2. Why the Obvious Solution Fails

Standard transfer-learning approaches, such as fine-tuning ResNet or Inception with a traditional softmax or sigmoid classification head, were considered. On Commons, they face four critical limitations: **Fixed label space.** A new category forces full retraining. **Our answer:** categories are retrievable objects in a shared image-text space. **Long tail.** Most categories have fewer than 20 images; supervised heads ignore them. **Our answer:** kNN, text prototypes, and graph-smoothed prototypes. **Missing labels.** Treating unlabelled data as negative penalises correct ancestors. **Our answer:** asymmetric loss with margin and hierarchy-aware decoding. **No uncertainty.** A standard classifier always answers, even when unsure, contrary to human-in-the-loop use. **Our answer:** conformal suggestion sets with explicit abstention.

3. Core Idea and Architecture

Treat categories as retrievable objects in a shared image-text embedding space, and make the system honest about what it knows.

Pipeline: Upload event → queue → frozen encoder → four scorers → graph-smoothed prototypes → calibrated fusion → hierarchy decoding → conformal filter → store suggestions and evidence. It is fully asynchronous and never blocks the upload.

Flow: Upload/queue → frozen SigLIP or CLIP → 4 scorers → calibrated fusion → hierarchy decode → conformal set or abstain → explain and review.

3.1 Feature Extraction

A pre-trained open-weight vision-language encoder (SigLIP/CLIP) produces L2-normalised embeddings; it is frozen by default for efficiency and stability, with the last blocks optionally unfrozen, serving as the foundational feature extractor.

3.2 Four Complementary Scorers

kNN retrieval: Finds the K most similar already-categorised images; votes with temperature-softmax weights; neighbours become the explanation. *Covers:* head and torso categories, plus explanations. **Text prototype:** Cosine similarity between image and category-name embedding (with aliases and multilingual labels); zero-shot. *Covers:* new and rare categories. **Supervised head:** Per-category sigmoid on the embedding, trained with asymmetric loss and margin; weights initialised from text embeddings. *Covers:* frequent categories. **Metadata scorer:** Text model over title/description plus geotag and date features; feeds the fusion step, not used alone. *Covers:* events, places, abstract concepts.

Graph-smoothed category prototypes (the creative part): Each prototype blends its text embedding, the centroid of its image embeddings, and the average of its parent and sibling prototypes, so data-sparse tail categories borrow strength from their hierarchical neighbourhood without retraining.

Calibrated fusion: A logistic-regression stacker combines the four scorers, fitted on a held-out split; it runs only on the union of each scorer's top candidates (a few hundred), not on the whole label space.

Hierarchy-aware decoding: (1) If a category is selected, suppress its ancestors; (2) if an ancestor scores high but no descendant is confident, keep it. This turns the unlabelled-ancestor bias into a controlled decoding rule.

Conformal suggestion sets: Split conformal prediction on a held-out calibration set replaces a fixed top-k: for error rate α the set contains at least one correct category with probability $\geq 1-\alpha$. Confident images get short sets, ambiguous ones broader sets, unknown ones an explicit abstention ("manual categorisation required").

Interpretability: Every suggestion shows the nearest-neighbour images and text-match score behind it, so editors can audit and approve quickly (Wikimedia transparency principle).

4. Data and Training Protocol

Sources: Commons dumps and MediaWiki API (categorylinks, page, imageinfo); open-licence files only. **Cleaning:** Drop maintenance, tracking and process categories; resolve redirects; supervised head only for categories with $\geq$ 20–50 images; rarer ones served by retrieval and text. **Splits:** By upload timestamp (train earlier, test later); remove near-duplicates by embedding cosine threshold or perceptual hash; group by uploader to prevent style leakage. **Loss and feedback:** Asymmetric loss with probability margin tolerates missing labels; accepted suggestions become positives, re-

jected ones hard negatives.

5. Robustness to Quality and Resolution

Fixed-size resize at inference. Training augmentation: Gaussian blur, random down/up-scaling, JPEG re-encoding, random-resized crop. All metrics reported per resolution band.

6. Serving and Scale

10^4–10^5 uploads/day ($\approx$ 1 image/s at the upper end); one GPU or quantised CPU inference is enough. Model cost is reduced by distillation or INT8/ONNX/TensorRT after measuring accuracy loss. **Index:** 10^8 images $\times$ 512-d fp16 $\approx$ 102 GB flat; product quantisation at 64 B/vector $\approx$ 6.4 GB; HNSW or IVF-PQ queries take milliseconds.

7. Continual Learning

New images: Append embeddings to the index, no retraining. **New categories:** Add text embedding and any examples; usable immediately; supervised head extended at the next scheduled retrain. **Editor feedback:** Accept/reject events update the index and fusion weights. **Monitoring:** Acceptance rate and calibration per category family; recalibrate the conformal threshold and retrain on drift.

8. Evaluation Plan and Ablations

Single-label accuracy does not apply (many labels per image, huge label space), so we report:

- **Ranking metrics:** Precision@k, Recall@k, F1@k (sample-wise for user view, macro for tail view), mAP, and hierarchy-aware precision where an ancestor or descendant of a true label counts as partially correct.
- **Stratification:** Head / torso / tail; object / scene / event / abstract; resolution band; label language. Strict vs. human-reviewed score, since stored labels are incomplete.
- **Trust and systems:** Conformal coverage vs. target, average set size, abstention rate; latency (p50/p95), throughput, index size, memory.
- **Ablations for judges:** (1) kNN only, text only, head only, full fusion; (2) with/without graph-smoothed prototypes (tail recall); (3) with/without hierarchy decoding; (4) fixed top-k vs. conformal sets.

9. Mapping to the Likely Judging Rubric

Weights are our estimates, not the organisers'.

Criterion (weight)	How CommonsLens answers it
Accuracy on new images (40%)	Temporal hold-out, fusion of four scorers, tail-aware metrics
Efficiency and scalability (15%)	Frozen encoder, PQ index, asynchronous pipeline
Category breadth (10%)	Text and metadata scorers for events and abstract concepts
Robustness (10%)	Degradation augmentation, resolution-stratified metrics
Continual learning (10%)	Index append, new-category path, feedback loop
Wikimedia fit (10%)	Open weights, neighbour explanations, human-in-the-loop, regional audit
Presentation (5%)	Live demo with explanations and abstention; honest limitations

10. Risks and Mitigations

Risk	Mitigation
Train/test leakage from near-duplicates	Temporal split, perceptual-hash and embedding dedup, uploader grouping
Missing labels treated as negatives	Asymmetric loss with margin; reviewed-sample evaluation; reject events as true negatives
Model learns process categories	Remove maintenance and hidden process categories before training
Geographic and cultural bias	Stratified reporting by region and language; multilingual labels; sampling not driven only by upload volume
Sensitive content and people	No identification of individuals; content categories only; human approval required